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Principles of Macroeconomics

Lecture 3: Production and Growth

Prof. Dr. Jan-Egbert Sturm

SCHWEIZERISCHE NATIONALBANK
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SWISS NATIONAL BANK

Karl Brunner Distinguished Lecture

"Inflation" by John H. Cochrane,
Hoover Institution (Stanford University)

Thursday, 2 October 2025, ETH Zurich, Audi Max (HG F 30)
Drinks reception: 4.45 – 5.30 pm; Lecture: 5.30 – 7.00 pm

Welcome by Joël Mesot
President of ETH Zurich

Introduction by Martin Schlegel
Chairman of the Governing Board, Swiss National Bank

Moderated by Tanya König

Live streaming www.snb.ch, Research TV

Please note that there will be a security check at the entrance to the lecture hall. Please arrive well in advance of the start of the event and refrain from bringing bags that are larger than A4 (21 x 30 cm).

Hosted by **KOF** Swiss Economic Institute and Chair of Macroeconomics: Innovation and Policy

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SFSO Video: What have statistics ever done for us?

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Principles of Macroeconomics

Lecture 2: National Accounting

Prof. Dr. Jan-Egbert Sturm

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- *Gross domestic product* (GDP) is a measure of the income and expenditures of an economy.

- Because for the total economy income equals expenditure,
  we can calculate GDP by adding up total expenditure in the economy,
  or by adding up total income in the economy.

- **GDP is the total market value of all final goods and services produced within a country in a given period of time.**

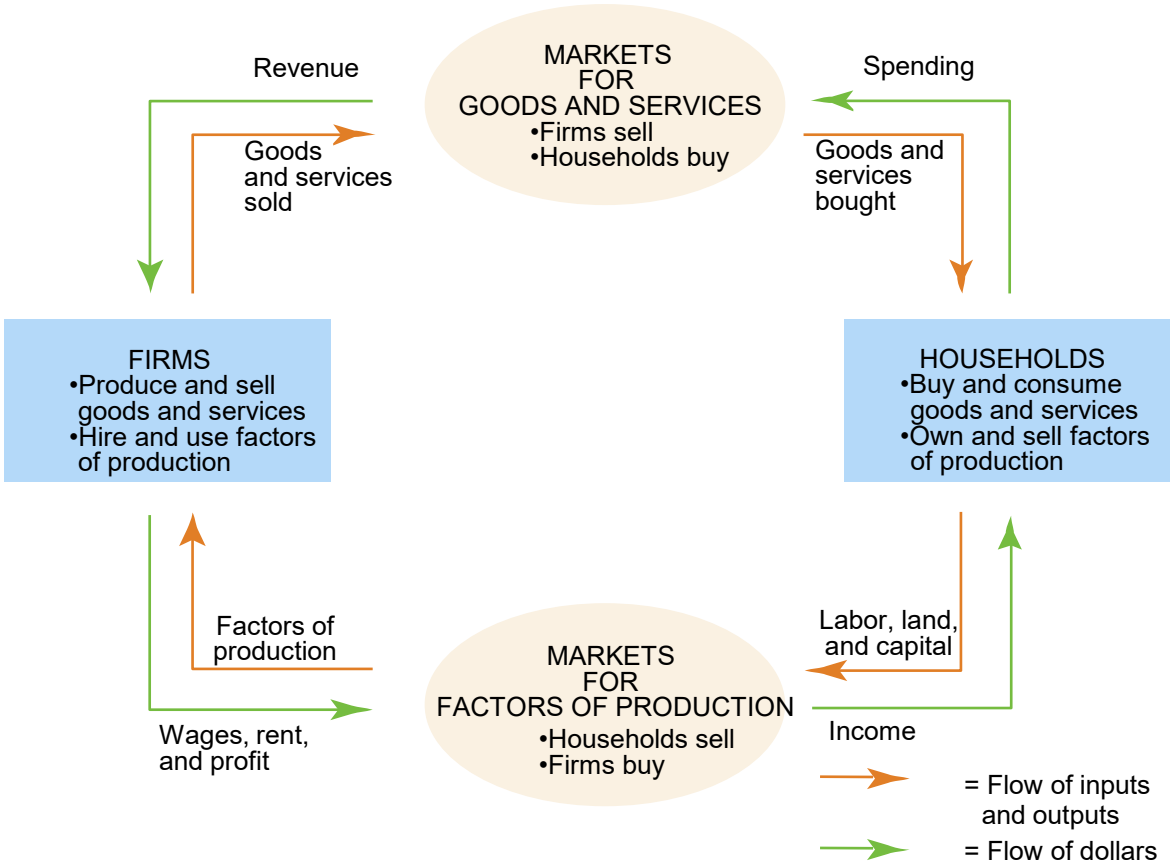
THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- "GDP is the Market Value . . ."
  - Output is valued at market prices.
- ". . . Of All. . ."
  - Includes all items produced in the economy and legally sold in markets
    - What Is Not Counted in GDP?
      - GDP excludes most items that are produced and consumed at home and that never enter the marketplace.
      - It often excludes items produced and sold illicitly, such as illegal drugs.
- ". . . Final . . ."
  - It records only the value of final goods and services, not intermediate goods (the value is counted only once).

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

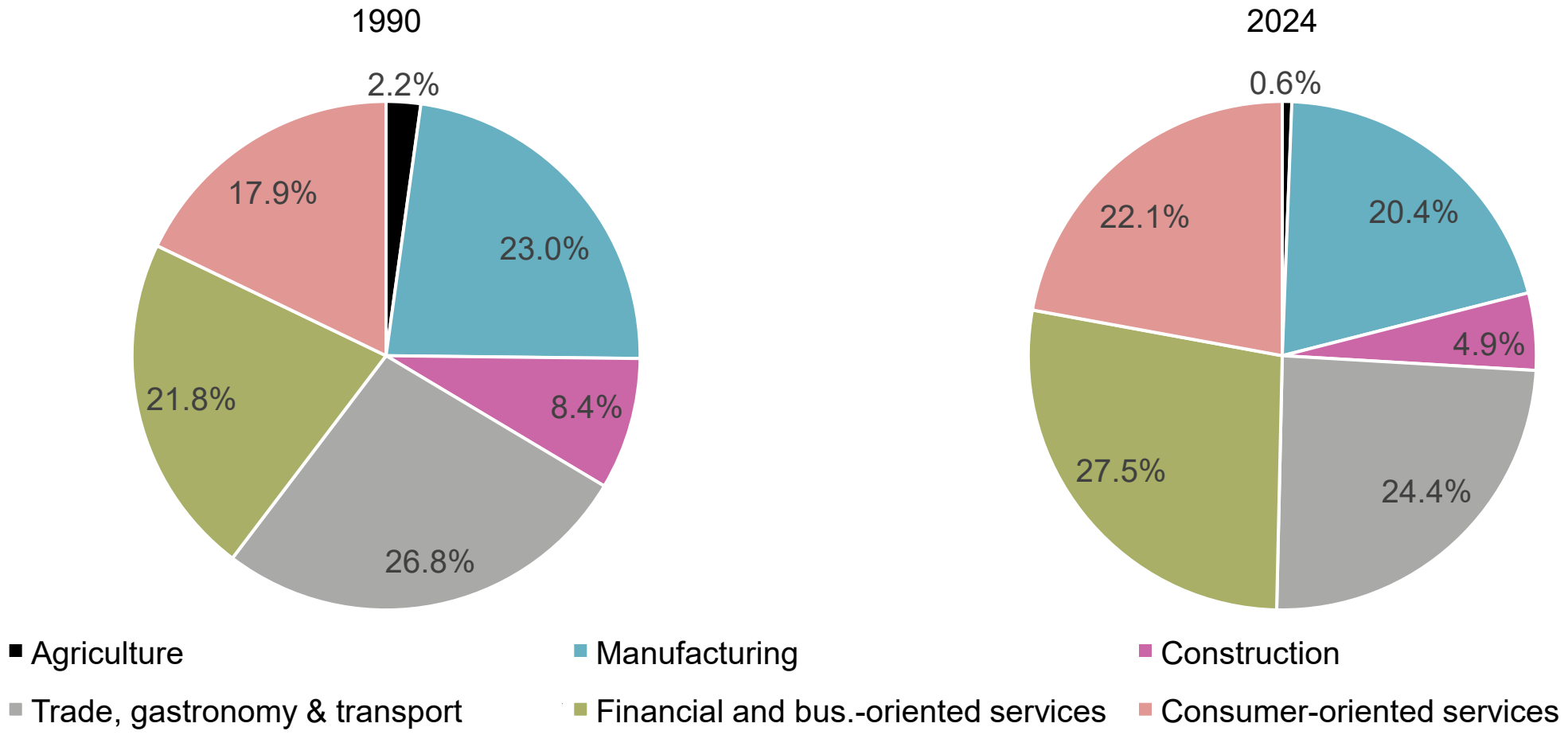
- “. . . Goods and Services . . .”
  - It includes both tangible goods (food, clothing, cars) and intangible services (haircuts, housecleaning, doctor visits).
- “. . . Produced . . .”
  - It includes goods and services currently produced, not transactions involving goods produced in the past.
- “ . . . Within a Country . . .”
  - It measures the value of production within the geographic confines of a country.
- “. . . In a Given Period of Time.”
  - It measures the value of production that takes place within a specific interval of time, usually a year or a quarter (three months).

How to measure GDP? Three main approaches



1. Expenditure Approach (demand side)
   - GDP = consumption + investment + government spending + net exports
2. Production Approach (supply side)
   - GDP = ∑(Gross Output− Intermediates)
3. Income Approach (income side)
   - GDP = Wages+Rents+Interest+Profits

Swiss GDP from the production (supply) side



Source: SFSO (19/09/2025)

DEMAND-SIDE COMPONENTS OF GDP (Y)

- Consumption (C)
  - The spending by households on goods and services, with the exception of purchases of new housing
- Investment (I)
  - The spending on capital equipment, inventories, and structures, including new housing
- Government Purchases (G)
  - The spending on goods and services by local, state, and federal governments
    - Does not include transfer payments, as these are not made in exchange for currently produced goods or services
- Net Exports (NX = X-M)
  - Exports (X) minus imports (M)

$$Y = C + I + G + NX$$

$$= C^{dom} + C^{for} + I^{dom} + I^{for} + G^{dom} + G^{for} + X - (C^{for} + I^{for} + G^{for})$$

$$= C^{dom} + I^{dom} + G^{dom} + X$$

GDP and its demand components (Swiss GDP in 2024: 826 Bln CHF (in prices of 2023))

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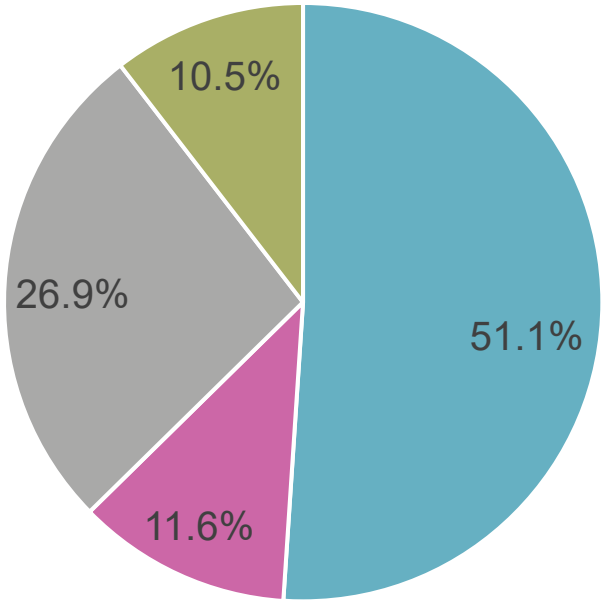
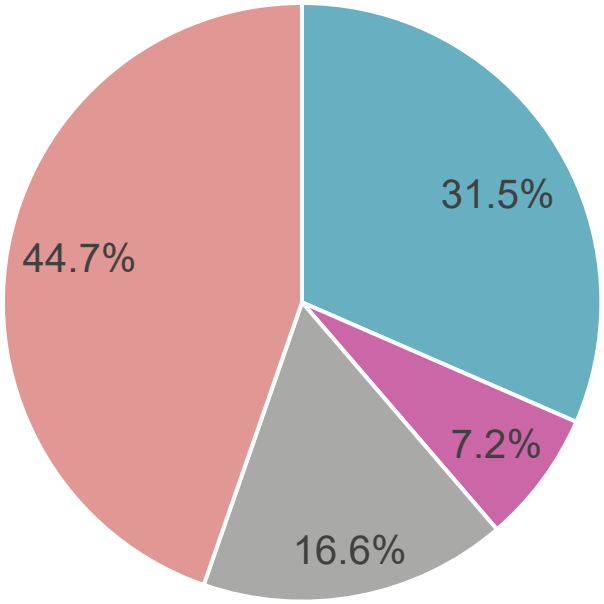
Gross Domestic Product:
$Y = C + G + I + (X-M)$



Total demand: $D = C + G + I + X$
$D = (C^{dom}+C^{for}) + (G^{dom}+G^{for}) + (I^{dom}+I^{for}) + X$



C = Private consumption G = Public consumption I = Investment NX = Net exports X = Exports

Source: SFSO (19/09/2025)

(nominal) GDP and Its Components: Switzerland

	Total (in Mln. CHF)	Per person (in CHF)	Percent of total
	2024		
GDP, Y	825,796	91,687	100%
Consumption, C	421,716	46,823	51%
Investment, I	223,114	24,772	27%
Government consumption, G	95,813	10,638	12%
Net exports, NX = X-M	85,153	9,454	10%
Exports, X	596,944	66,278	72%
Imports, M	511,791	56,824	62%

Source: SFSO (19/09/2025)

(nominal) GDP and Its Components: Switzerland

	Total (in Mln. CHF)	Per person (in CHF)	Percent of total
	2024		
GDP, Y	853,957	94,814	100%
Consumption, C	421,087	46,753	49%
Investment, I	228,223	25,339	27%
Government consumption, G	101,745	11,297	12%
Net exports, NX = X-M	102,902	11,425	12%
Exports, X	608,037	67,510	71%
Imports, M	505,135	56,085	59%

Source: SFSO (29/09/2025)

REAL VERSUS NOMINAL GDP

- If total spending rises from one year to the next, this can be because...
  - ...the economy is producing a larger output of goods and services *and/or*
  - ...goods and services are being sold at higher prices.
- We want a measure of the total output of goods and services that is not affected by changes in the prices of those goods and services.

- *Nominal GDP* values the production of goods and services at *current prices*.
- *Real GDP* values the production of goods and services at *constant prices*.

- Real GDP is a more accurate measure of economic well-being than nominal GDP, because changes in real GDP are not affected by changes in prices, but reflect only changes in production.

Real and Nominal GDP

Prices and Quantities				
Year	Price of Hot dogs	Quantity of Hot dogs	Price of Hamburgers	Quantity of Hamburgers
2005	$1	100	$2	50
2006	$2	150	$3	100
2007	$3	200	$4	150

Calculating Nominal GDP	
2005	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2006	($2 per hot dog × 150 hot dogs) + ($3 per hamburger × 100 hamburgers) = $600
2007	($3 per hot dog × 200 hot dogs) + ($4 per hamburger × 150 hamburgers) = $1,200

Calculating Real GDP (base year 2005)	
2005	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2006	($1 per hot dog × 150 hot dogs) + ($2 per hamburger × 100 hamburgers) = $350
2007	($1 per hot dog × 200 hot dogs) + ($2 per hamburger × 150 hamburgers) = $500

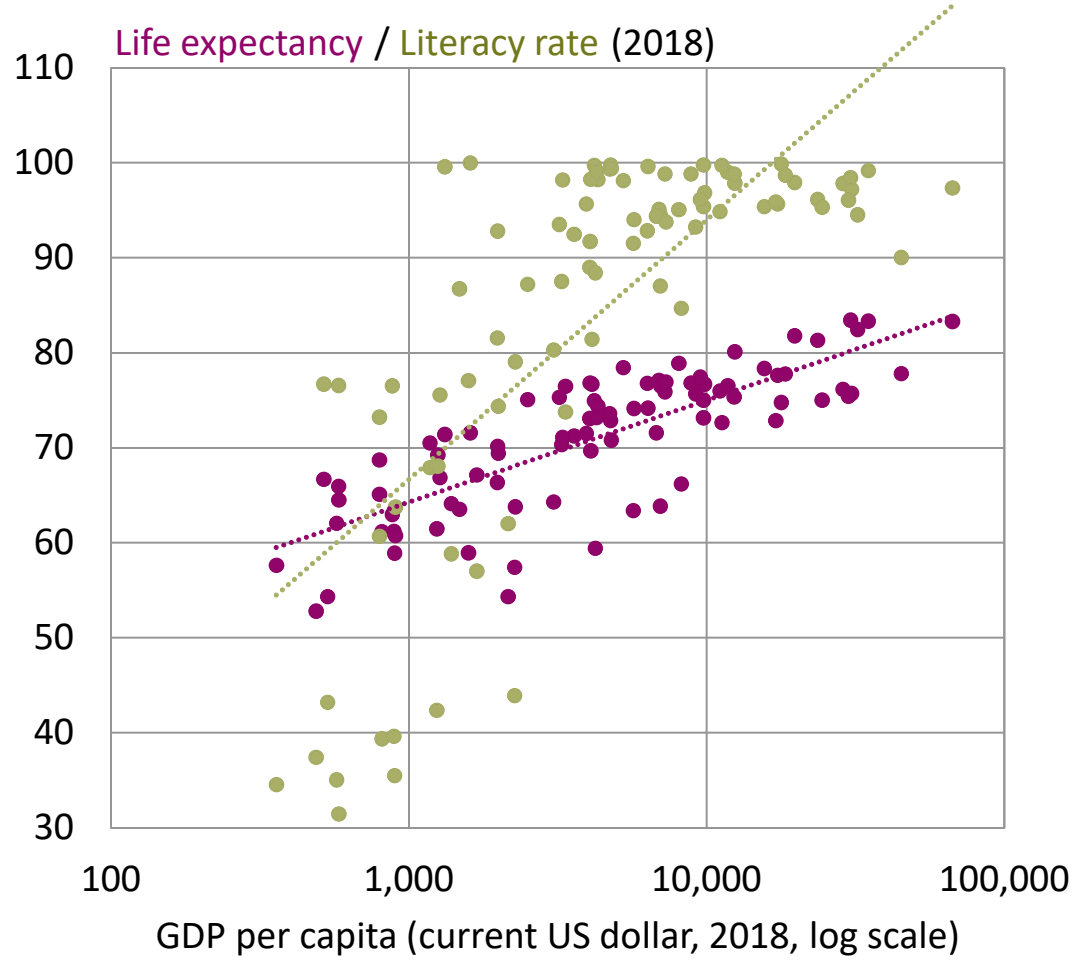
Measurement Issues in Practice

- In most western countries the statistical offices produce a single measure of GDP and they do this by combining income, expenditure and output approaches.
  - In theory these measures should produce the same result (in practice they don't).

- There are various practical difficulties in measuring GDP.
  - Under-reporting in order to avoid tax
  - Changes in the goods and services produced over time *render the use of base years problematic*.

- Annual chain linking can be used to tackle this latter problem.
  - Rather than updating the base year every five or ten years,
    this method does it every year, calculating the prices in previous years' prices.

IS GDP A GOOD MEASURE OF ECONOMIC WELL-BEING?



Sources: IMF, World Bank

- GDP is the best single measure of economic well-being
- GDP per person tells us the income and expenditure of the average person in the economy
- Higher GDP per person indicates a higher standard of living
- GDP is not a perfect measure of happiness or quality of life
  - See article (Looking Beyond GDP) on moodle
- Some things that contribute to well-being are not included
  - The value of leisure
  - The value of a clean environment
  - The distribution of income
  - The value of almost all activity that takes place outside of markets, such as the value of the time parents spend with their children and the value of volunteer work

- Income, life expectancy and CO2 emission (video):
- The MONET 2030 indicator system gives an overview of sustainable development in Switzerland and goes beyond GDP

MEASURING THE COST OF LIVING

- The *inflation rate* is the percentage change in the price level from the previous period.

  Formally: $$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} * 100$$

- *Inflation* refers to a situation in which the economy's overall price level is rising.

- *Deflation* refers to situation in which the economy's overall price level is falling.

The GDP Deflator

- The *GDP deflator* is a measure of the price level calculated as the ratio of nominal GDP to real GDP times 100.
- It tells us what portion of the rise in nominal GDP is attributable to a rise in prices rather than a rise in the quantities produced.
- The GDP deflator is calculated as follows

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

	Calculating the GDP Deflator
2005	($200/$200) × 100 = 100
2006	($600/$350) ×100 = 171
2007	($1,200/$500) × 100 = 240

THE CONSUMER PRICE INDEX

- The *consumer price index (CPI)* is a measure of the overall cost of the goods and services bought by a typical consumer.

- It is used to monitor changes in the cost of living over time.

- When the CPI rises, the typical consumer has to spend more money to maintain the same standard of living.

How the Consumer Price Index Is Calculated

1. *Fix the basket*: Determine what prices are most important to the typical consumer.
   - Identify a market basket of goods and services the typical consumer buys.
   - Consumer surveys to set the weights for the prices of those goods and services.
2. *Find the prices*: Find the prices of each of the goods and services in the basket for each point in time.
3. *Compute the basket's cost*: Use the data on prices to calculate the cost of the basket of goods and services at different times.
4. *Choose a base year and compute the index:*
   - Designate one year as the base year, making it the benchmark against which other years are compared.
   - Compute the index by dividing the price of the basket in one year by the price of the basket in the base year.

$$\text{Consumer price index in year t} = \frac{\text{Price of basket in year t}}{\text{Price of basket in base year}} \times 100$$

5. *Compute the consumer price inflation rate:* The *consumer price inflation rate* is the percentage change in the price level – as measured by CPI – from the preceding period.

$$\text{Inflation rate in year 2} = \frac{\text{CPI in year 2 - CPI in year 1}}{\text{CPI in year 1}} \times 100$$

Calculating Consumer Price Index and the Inflation Rate: An Example

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

Basket = 4 hot dogs, 2 hamburgers

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2005	$1	$2
2006	2	3
2007	3	4

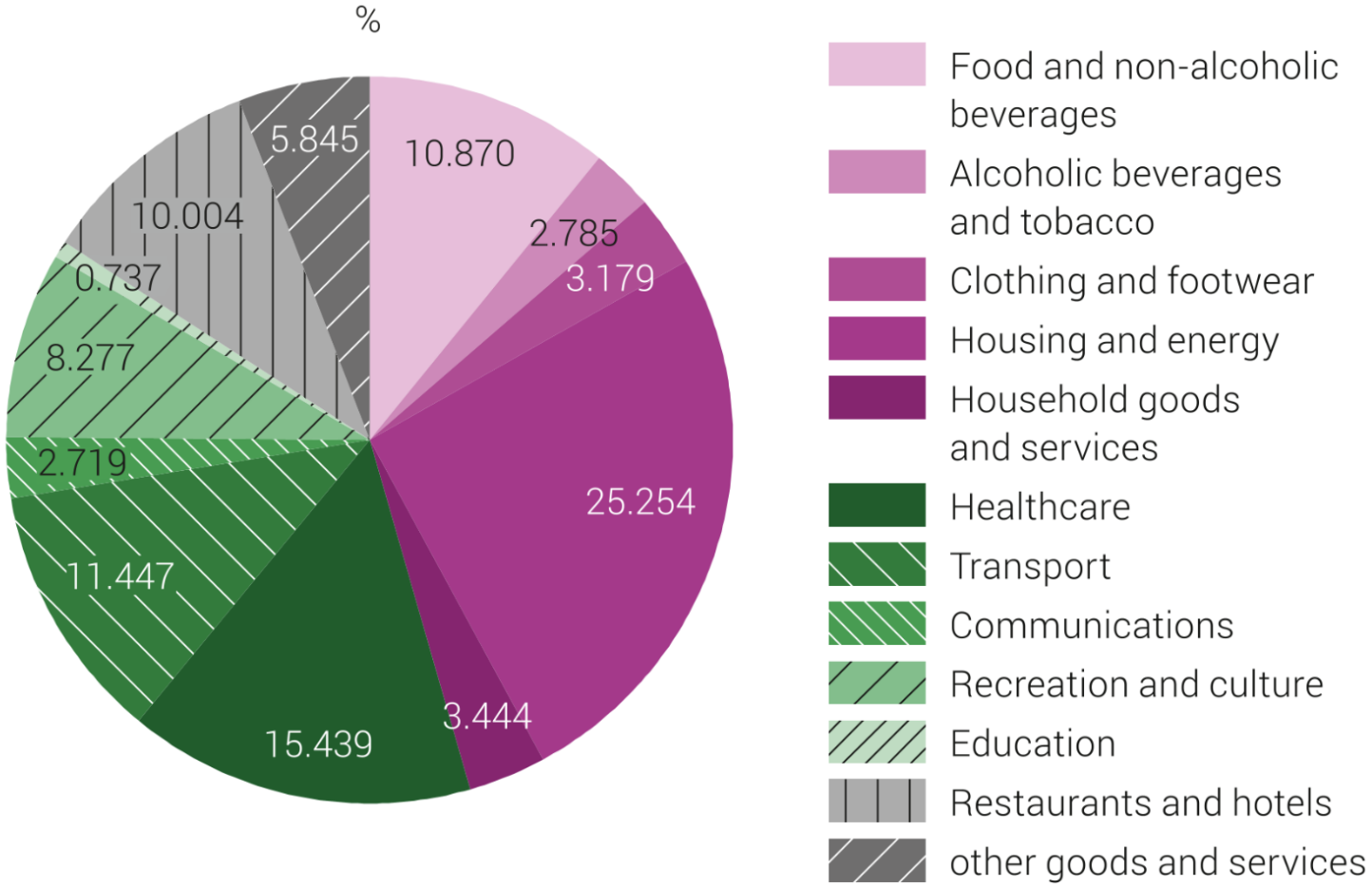
Calculating Consumer Price Index and the Inflation Rate: An Example

Step 3: Compute the Cost of the Basket of Goods in Each Year	
2005	($1 per hot dog × 4 hot dogs) + ($2 per hamburger × 2 hamburgers) = $8 per basket
2006	($2 per hot dog × 4 hot dogs) + ($3 per hamburger × 2 hamburgers) = $14 per basket
2007	($3 per hot dog × 4 hot dogs) + ($4 per hamburger × 2 hamburgers) = $20 per basket

Step 4: Choose One Year as a Base Year (2005) and Compute the Consumer Price Index in Each Year	
2005	($8/$8) × 100 = 100
2006	($14/$8) × 100 = 175
2007	($20/$8) × 100 = 250

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year	
2006	(175 − 100)/100 × 100 = 75%
2007	(250 − 175)/175 × 100 = 43%

CPI basket and weights in Switzerland, 2024



Source: SFSO

Historical developments in RPI (≈CPI) weights for the United States

RPI GROUP	1987	2012
Food	167	114
Catering	46	47
Alcohol	76	56
Tobacco	38	29
Housing	157	237
Fuel and light	61	46
Household goods	73	62
Household services	44	67
Clothing and footwear	74	45
Personal goods and services	38	39
Motoring expenditure	127	131
Fares and other travel costs	22	23
Leisure goods	47	33
Leisure services	30	71

SFSO videos on the most important price statistics in Switzerland

- Price statistics in brief
  - https://www.youtube.com/watch?v=xZdopkGNdeo

- The Consumer Price Index (CPI) at a glance
  - https://www.youtube.com/watch?v=t2BXjbiLmMo

- How does the consumer price index work?
  - https://www.youtube.com/watch?v=lhmjO-gvo9s

- The Producer Price Index (PPI) at a glance
  - https://www.youtube.com/watch?v=7FvQW583sew

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
  2. Introduction of new goods
  3. Unmeasured quality changes

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
     - The basket does not change to reflect consumer reaction to changes in relative prices.
       - Consumers substitute toward goods that have become relatively less expensive.
       - The index overstates the increase in cost of living by not considering consumer substitution.
  2. Introduction of new goods
  3. Unmeasured quality changes

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
  2. Introduction of new goods
     - The basket does not reflect the change in purchasing power brought on by the introduction of new products.
       - New products result in greater variety, which in turn makes each euro more valuable.
       - Consumers need less money to maintain any given standard of living.
  3. Unmeasured quality changes

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
  2. Introduction of new goods
  3. Unmeasured quality changes
     - If the quality of a good rises from one year to the next, the value of a euro rises, even if the price of the good stays the same.
       - If the quality of a good falls from one year to the next, the value of a euro falls, even if the price of the good stays the same.
       - The statistical office tries to adjust the price for constant quality, but such differences are hard to measure.

Problems in Measuring the Cost of Living

- The issue over accurate measurement is important because many government programs use the CPI to adjust for changes in the overall level of prices.

- The statistical office calculates other prices indexes:
  - The *producer price index*, which measures the cost of a basket of goods and services bought by firms rather than consumers.
    - Because firms eventually pass on higher costs to consumers in the form of higher prices on products, the producer price index is believed to be helpful in predicting changes in the CPI.

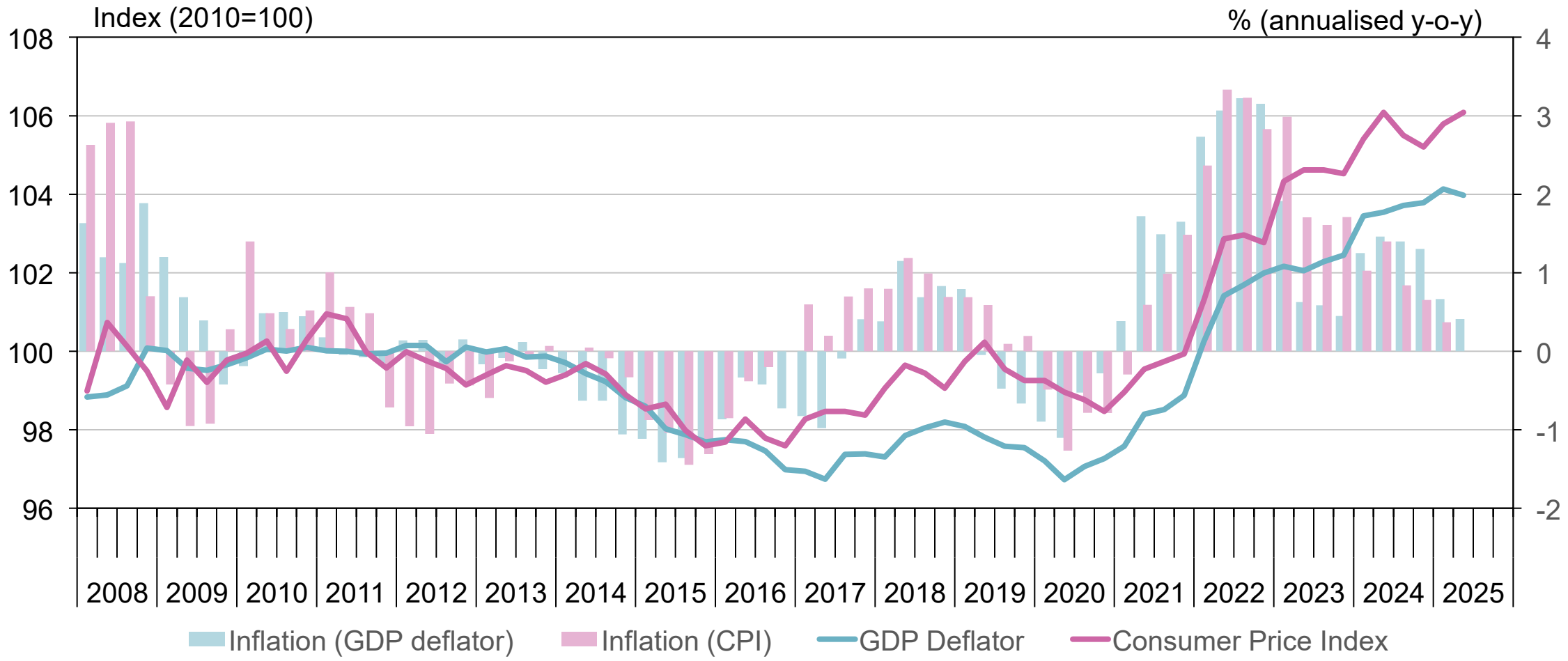
The GDP Deflator versus the Consumer Price Index

- Remember, the GDP deflator is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- Economists and policymakers monitor both the GDP deflator and the consumer price index to gauge how quickly prices are rising.
- There are two important differences between the indexes that can cause them to diverge.
  1. The *GDP deflator* reflects the prices of all goods and services *produced domestically*, whereas... …the *consumer price index* reflects the prices of all goods and services *bought by consumers*.
  2. The *consumer price index* compares the price of a *fixed basket* of goods and services to the price of the basket in the base year (only every five years does the SFSO change the basket)... …whereas the *GDP deflator* compares the price of *currently produced* goods and services to the price of the same goods and services in the base year.

Two measures of inflation in Switzerland



Sources: SFSO, Seco

CORRECTING ECONOMIC VARIABLES FOR THE EFFECTS OF INFLATION

- Price indexes are used to correct for the effects of inflation when comparing money figures from different times.
- Example:
  - In 1947 MPs earned £1,000
  - Assume in 2020 that MPs earned £74,000 but prices have risen since 1947.
  - CPI index in 1947 = 28.9
  - CPI index in 2020 = 1,018.6
- Do the following to convert (inflate) MPs' salary in 1947 to a figure in 2020 values:

$$\text{Salary}_{2020} = \text{Salary}_{1947} \frac{\text{Price level in 2020}}{\text{Price level in 1947}}$$

$$= £1{,}000 \frac{1{,}018.6}{28.9} = £35{,}246$$

Indexation

- When some money amount is automatically corrected for inflation by law or contract, the amount is said to be *indexed* for inflation.
  - Many government transfer payment systems use indexation for the benefits.
    The government often also adjusts tax brackets used for income tax in line with inflation.
  - There are uses of indexation in the private sector as well.
    Many contracts between firms and trade unions include Cost-of-Living Allowances.

Real and Nominal Interest Rates

- Interest represents a payment in the future for a transfer of money in the past.
- The *nominal interest* rate is the interest rate usually reported and not corrected for inflation.
  - It is the interest rate that a bank pays.
- The *real interest rate* is the nominal interest rate that is corrected for the effects of inflation.
- Here is a simple example
  - You borrowed €1,000 for one year.
  - Nominal interest rate was 15%.
  - During the year inflation was 10%.

$$\textit{Real interest rate} = \textit{Nominal interest rate} - \textit{Inflation}$$

$$= 15\% - 10\% = 5\%$$

SUMMARY

- Because every transaction has a buyer and a seller, the total expenditure in the economy must equal the total income in the economy.
- Gross domestic product (GDP) measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services.
- GDP is the market value of all final goods and services produced within a country in a given period of time.
- GDP is divided among four components of expenditure: consumption, investment, government purchases, and net exports.
- Nominal GDP uses current prices to value the economy's production. Real GDP uses constant base-year prices to value the economy's production of goods and services.
- The GDP deflator—calculated from the ratio of nominal to real GDP—measures the level of prices in the economy.
- GDP is a good measure of economic well-being because people prefer higher to lower incomes.
- It is not a perfect measure of well-being because some things, such as leisure time and a clean environment, are not measured by GDP.

SUMMARY

- The consumer price index shows the cost of a basket of goods and services in one year relative to the cost of the same basket in the base year.
- The index is used to measure the overall level of prices in the economy.
- The percentage change in the CPI measures the inflation rate.
- The consumer price index is an imperfect measure of the cost of living for the following three reasons:
  - substitution bias / the introduction of new goods / and unmeasured changes in quality.
- The GDP deflator differs from the CPI because it includes goods and services produced rather than goods and services consumed.
- In addition, the CPI uses a fixed basket of goods, while the GDP deflator automatically changes the group of goods and services over time as the composition of GDP changes.
- Money figures from different points in time do not represent a valid comparison of purchasing power.
- Various laws and private contracts use price indexes to correct for the effects of inflation.
- The real interest rate equals the nominal interest rate minus the rate of inflation.

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Principles of Macroeconomics

Lecture 3: Production and Growth

Prof. Dr. Jan-Egbert Sturm

Production and Growth

- A country's standard of living depends on its ability to produce goods and services
- Within a country there are large changes in the standard of living over time
  - In Switzerland (real) GDP per capita (average income) has grown on average by about 0.8 per cent per year since 2000
  - This means that GDP per capita increased by 21 per cent since 2000
  - It implies that GDP per capita doubles roughly every 87 years
- Annual growth rates that seem small become significant when compounded for many years
  - If the annual growth rate of Swiss GDP per capita had just been one-tenth of one percent higher since the start of this century, an average individual would have earned an additional CHF 24,590

Huge effects from small differences

Annual growth rate of income per capita	Percentage increase in living standards after		
	…25 years	…50 years	…100 years
0.5%	13.3%	28.3%	64.7%
1.0%	28.2%	64.5%	170.5%
1.5%	45.1%	110.5%	343.2%
2.0%	64.1%	169.2%	624.5%
2.5%	85.4%	243.7%	1081.4%

Production and Growth

- ***Productivity*** refers to the amount of goods and services produced for each hour of a worker's time
  - A nation's standard of living is determined by the productivity of its workers

- ***GDP per capita*** takes the level of real GDP at a point in time and divides it by the population to get a measure of income per head
- ***GDP per worker*** is a measure of the income per head of the working population

- Living standards, as measured by real GDP per capita, vary significantly among nations
  - The United Kingdom is an advanced economy. In 2019, its GDP per capita was $42,230
  - Poland is a middle-income country. In 2019, its GDP per capita was $15,630
  - The Central African Republic is a poor country. In 2019, its GDP per capita was only $441

Living Standards Around The World

- The United Kingdom is an advanced economy
  - In 2011, its GDP per person was $39,038
- Mexico is a middle-income country
  - In 2011, its GDP per person was $10,047
- Burundi is a poor country
  - In 2011, its GDP per person was only $271

Growth Theory

- The trend rate of growth is:
  - The average sustainable rate of economic growth over a period of time
- Trend growth and actual growth are important factors when we look at business cycles
- The Solow theory has been used to explain economic growth
  - It identifies the rate of human and physical capital and population growth as being key determinants
- Economists have identified other factors that may influence productivity and thereby economic growth
  - How open to trade a country is
  - How easy it is to do business and how and the extent to which corruption is minimised
  - The extent to which violence, war and conflict exist in a country
  - Regional, institutional and cultural characteristics
  - Geographical factors such as physical resource endowments and climate
- Productivity plays a key role in determining living standards for all nations in the world

Why Productivity Is So Important

- To understand the large differences in living standards across countries, we must focus on the production of goods and services
- A nation can enjoy a high standard of living if it can produce a large quantity of goods and services
- Hence, to understand the large differences in living standards we observe across countries or over time, we focus on the production of goods and services

How Productivity Is Determined

- The inputs used to produce goods and services are called the factors of production
- The factors of production directly determine productivity
- Labour productivity is the amount of output per worker
  - It will rise if the capital stock increases
  - However, because of diminishing marginal product, the rate of increase in average labour productivity will decline as the capital stock per worker increases
- The factors of production include:
  - Physical capital
  - Human capital
  - Natural resources
  - Technological knowledge
- The factors of production directly determine productivity

How Productivity Is Determined

- ***Physical capital*** per worker is the stock of equipment and structures that are used to produce goods and services
  - It includes machines and tools build or repair automobiles or furniture / Office buildings, schools, etc.
  - Physical capital is a produced factor of production
    - It is an input into the production process that in the past was an output from that process
- ***Human capital*** per worker is the economist's term for the knowledge and skills that workers acquire through education, training, and experience
  - Like physical capital, human capital raises a nation's ability to produce goods and services
- ***Natural resources*** are inputs used in production that are provided by nature
  - Renewable resources include trees and forests
  - Nonrenewable resources include petroleum and coal
    - Natural resources can be important but are not necessary for an economy to be highly productive
- ***Technological knowledge*** includes society's understanding of the best ways to produce
  - Human capital includes the resources expended transmitting this understanding to the labour force

FYI: The Production Function

- Economists often use a production function to describe the relationship between the quantity of inputs used in production and the quantity of output from production
- $Y = A\,F(L, K, H, N)$
  - Y = quantity of output
  - A = available production technology
  - L = quantity of labour
  - K = quantity of physical capital
  - H = quantity of human capital
  - N = quantity of natural resources
  - $F(\ )$ is a function that shows how the inputs are combined
- A production function has constant returns to scale if, for any positive number x,
  - $xY = A\,F(xL, xK, xH, xN)$
- A doubling of all inputs causes the amount of output to double as well
- Production functions with constant returns to scale have an interesting implication
- Setting $x = 1/L$ results in:
  - $Y/L = A\,F(1, K/L, H/L, N/L)$
  - Where:
    - Y/L = output per worker = labour productivity
    - K/L = physical capital per worker
    - H/L = human capital per worker
    - N/L = natural resources per worker

The Determinates Of Economic Growth

- The Solow growth model helps us understand some key elements of determinants of economic growth
  - GDP in any country can be assumed to be an extension of a firm's production function where the level of output is dependent on the factors of production employed
- Assumptions are
  - Two production factors: labour (L) and capital (K) (next to technology: A)
    - Output (GDP) = Y = AF(K,L)
  - Constant returns to scale: xY = AF(xK,xL)
    - This implies that capital and labour are each subject to ***diminishing marginal returns***
  - A closed economy and no government (G=0 & NX=0)
    - Therefore, Y = C + I + G + NX simplifies to Y = C + I

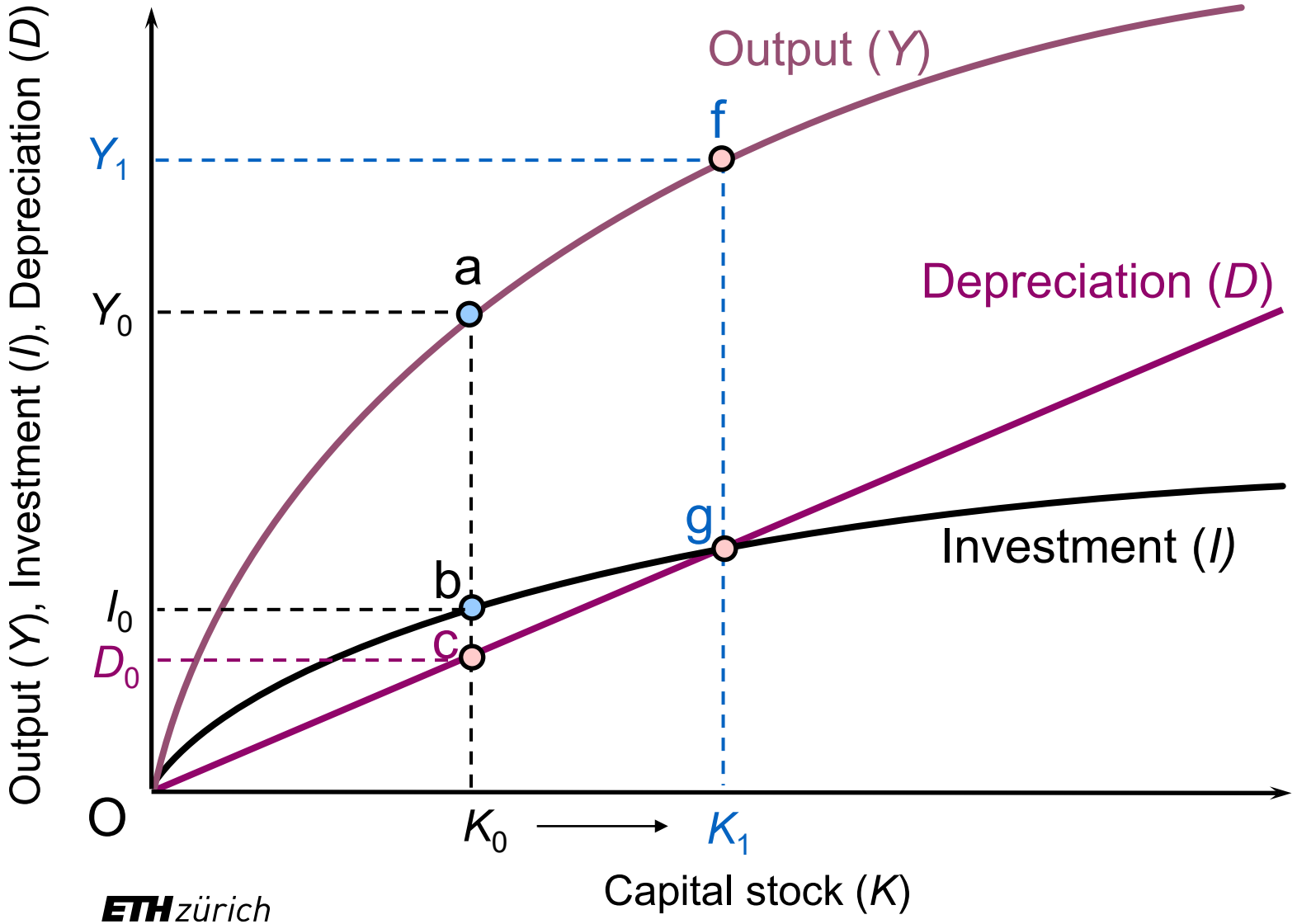
The Meaning of Saving and Investment

- For the economy as a whole, saving must be equal to investment:

$$(Y - C) \equiv S = I$$

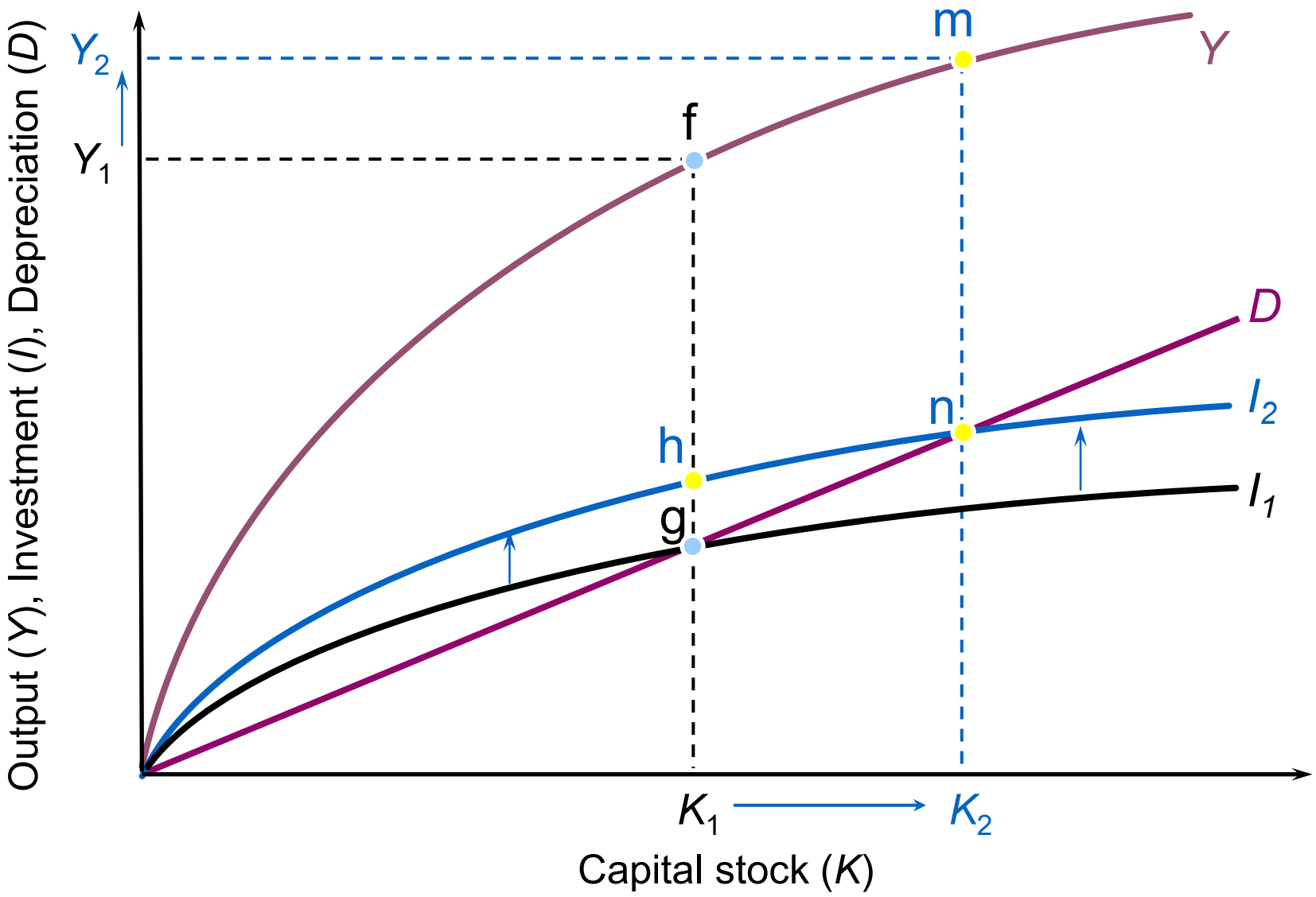
  - Note: “Investment” means the creation of new productive assets by firms and households, e.g. building a ***new*** factory or a ***new*** house
  - Buying an old house or a share in an existing company is only a financial transaction, although this is also often called “investment”
- Assume a constant share of income is saved and therefore invested: $S = sY$
- Firms invest in order to raise future productivity, or to replace depreciated capital
  - Assumption: capital depreciates with a constant rate: $D = \delta K$
- The difference between total investment (I) and depreciation (D) is called net investment

Investment, Growth, and Output

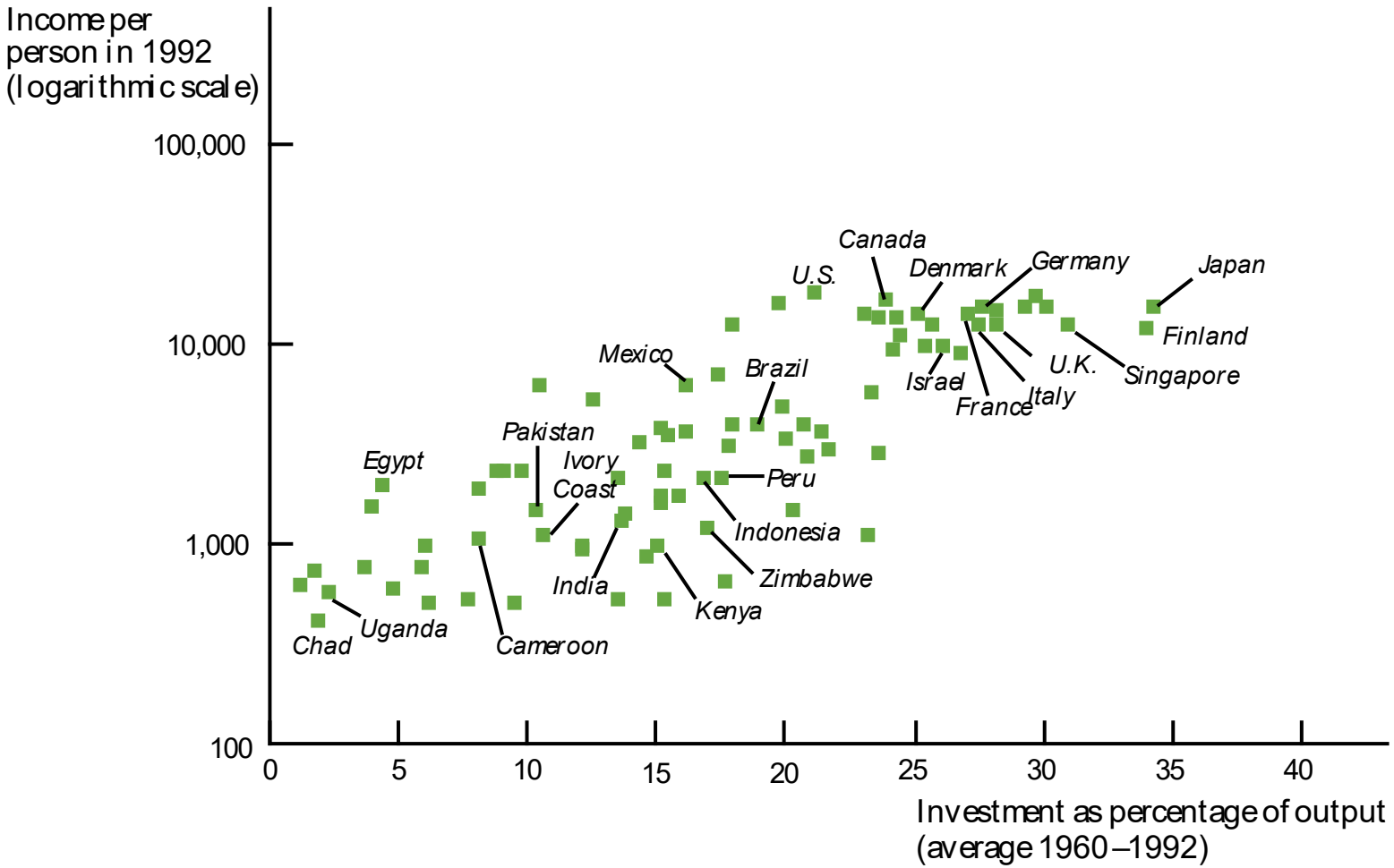


- For given A and L, increasing K is associated with rising GDP
- Diminishing marginal product of K implies with higher K lower GDP growth
- The distance between GDP and the investment level is consumption
- At a given saving rate, with positive net investment, the capital stock (and hence output) will grow until *D*=*I*
- ***Steady-state equilibrium*** is the point in a growing economy where investment is the same as spending on depreciation and the capital-output ratio remains constant
- Increasing the saving rate, allows K and Y to grow further, again until *D*=*I*

Effect of increase in the savings rate



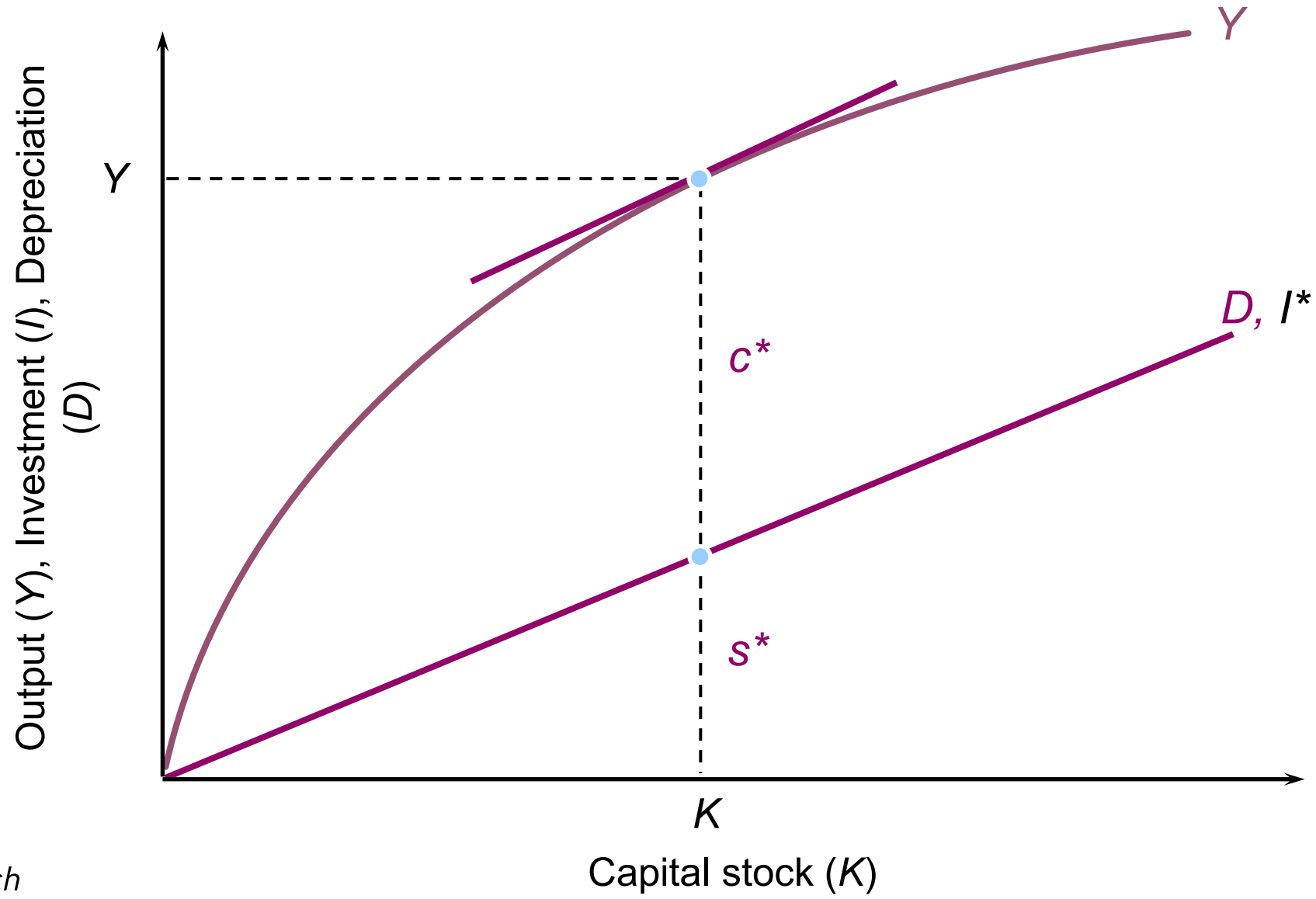
International Evidence on Investment Rates and Income per Person



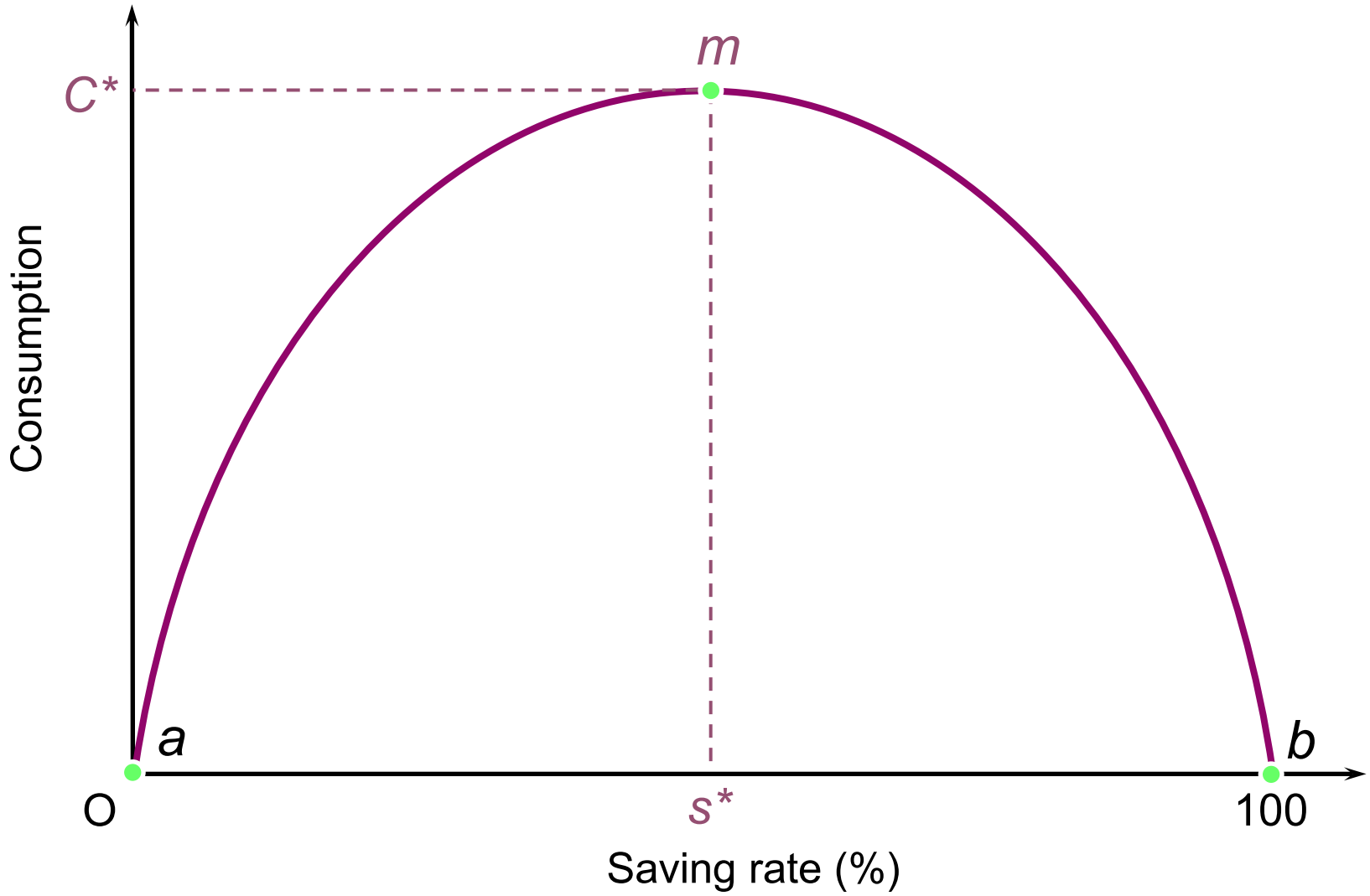
Investment, Growth and Output

- The economy's output is used for saving (investment) or consumption
- Long-run (steady state) consumption is the difference between steady state output and steady state depreciation (investment)
- There's a unique saving rate that maximizes the difference between output and investment

Steady state consumption



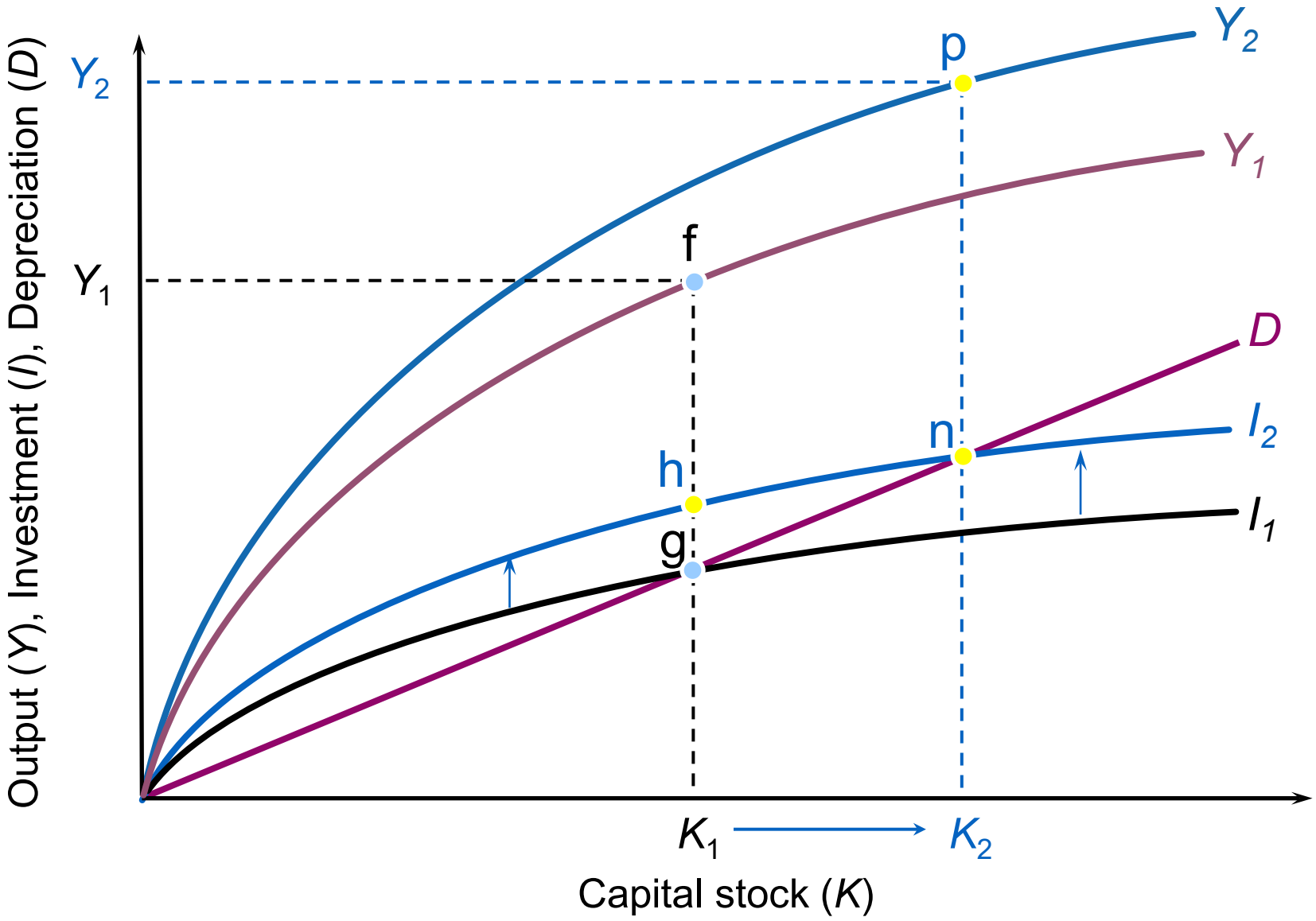
An optimum saving rate



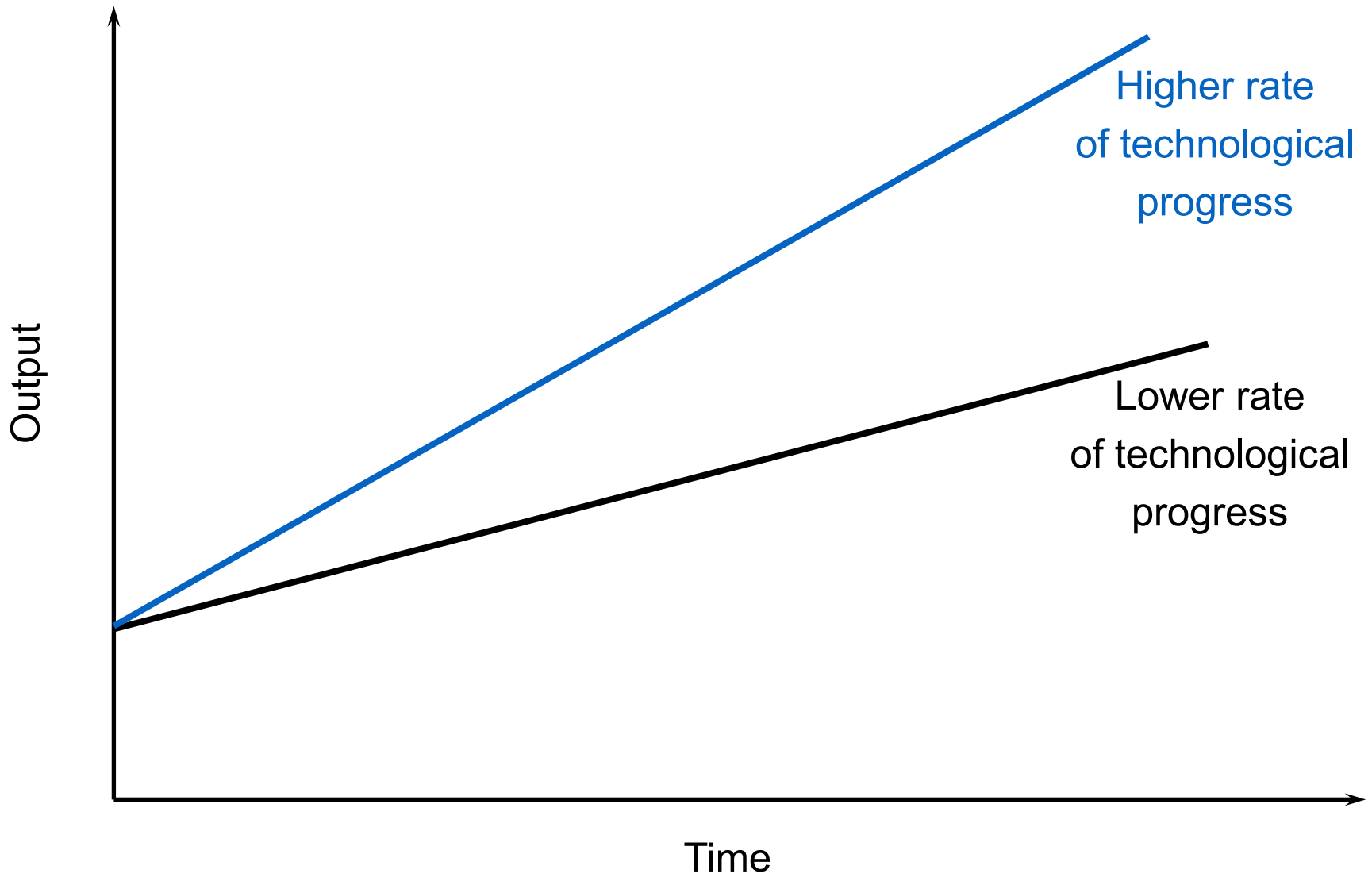
Technological Progress

- So far, the growth rate in the steady state is zero (A is constant)
- This is no longer the case if we allow for technological progress

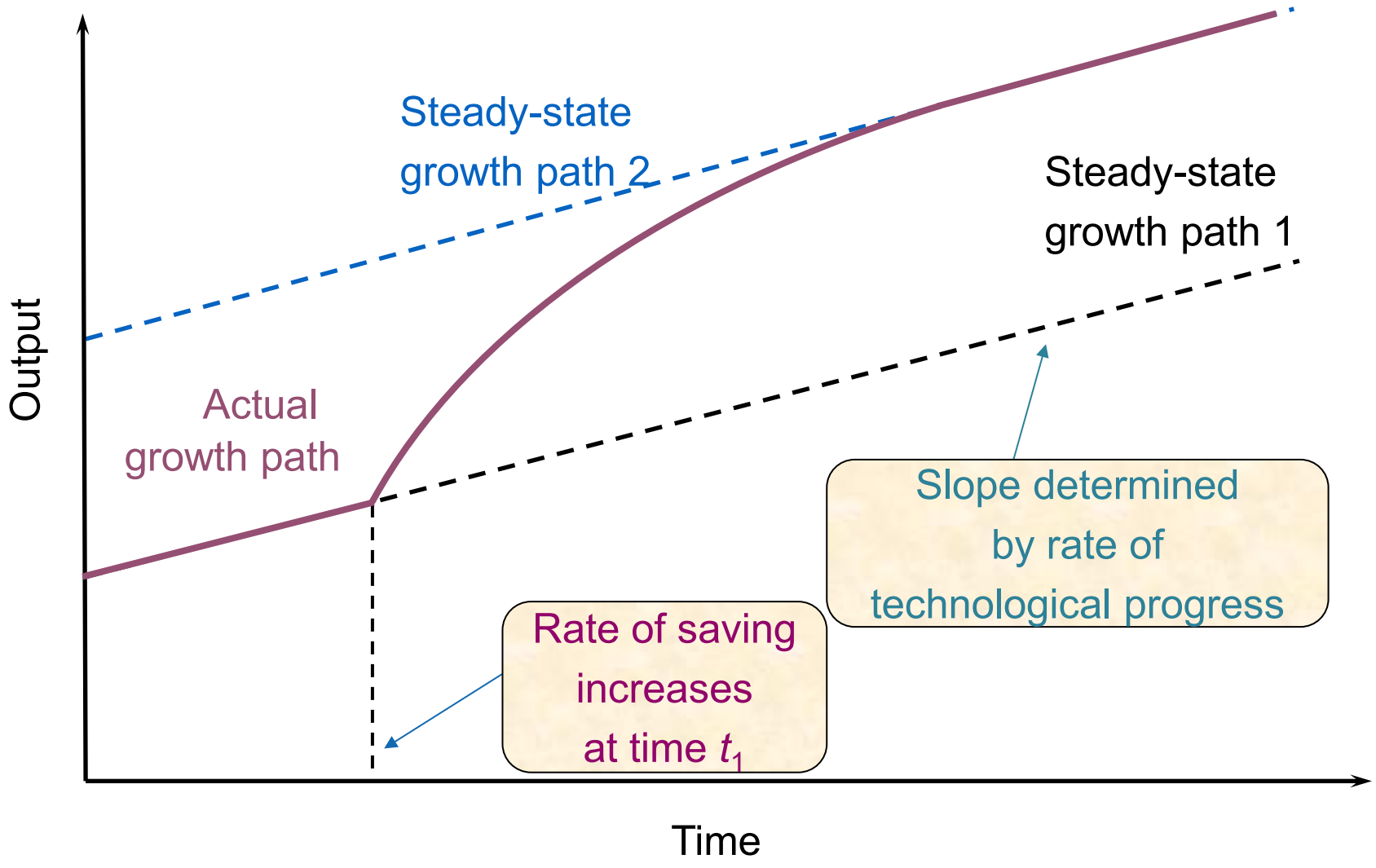
Effect of technological progress



Effect of technological progress



Effect of an increase in the saving rate with given technological progress



Long-run Equilibrium

- For an economy with a given level of physical capital, the growth path will be dependent on the:
  - Level of technology
  - Productivity of labour and capital
  - Savings rate which determines investment in physical capital
- The ratio of capital to labour is one aspect of how productive labour can be
- Countries like Korea and Vietnam have a supply of relatively cheap labour and have invested in capital and as a result labour productivity is relatively high and there is strong economic growth
- In many African countries investment in capital stock is extremely low and so labour productivity is also very low and there is weak economic growth
- The Solow model provides a means of understanding the transition of economies over time
  - Positive net investment in capital will increase capital per worker and lead to growth
  - However, investment is determined by the savings ratio
    - In less developed countries this may be relatively low because incomes are low
  - Also it is difficult to do business in these countries and therefore do not reach the steady state

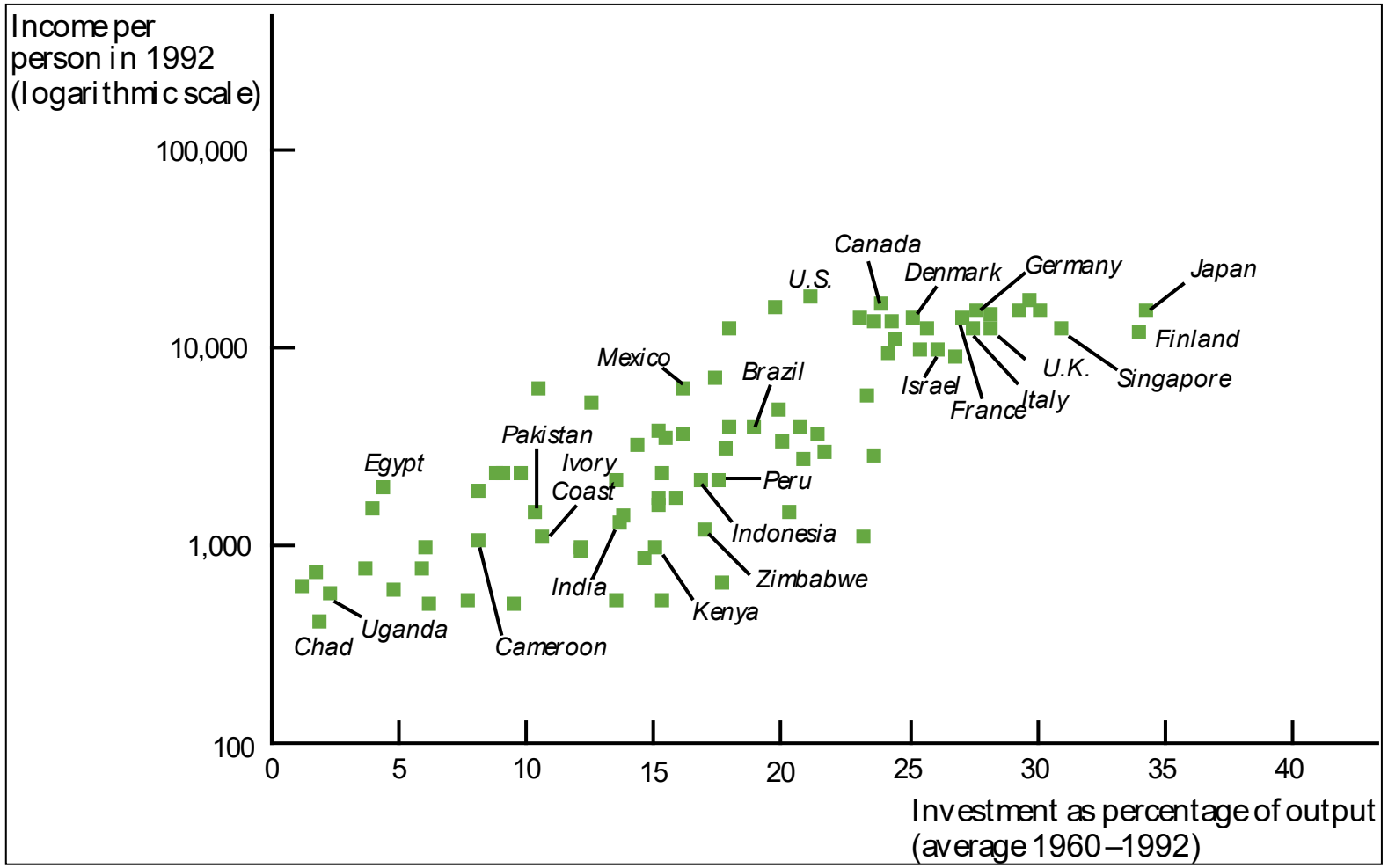
ECONOMIC GROWTH AND PUBLIC POLICY

- Government policies that raise productivity and living standards
  - Encourage saving and investment
  - Encourage investment from abroad
  - Encourage education and training
  - Establish secure property rights and maintain political stability
  - Promote free trade
  - Promote research and development

Saving and Investment

- One way to raise future productivity is to invest more current resources in the production of capital
- Since resources are scarce, producing more capital requires producing fewer consumption goods
- Reducing consumption = increasing saving
  This extra saving funds the production of investment goods
- Hence, a trade-off between current and future consumption

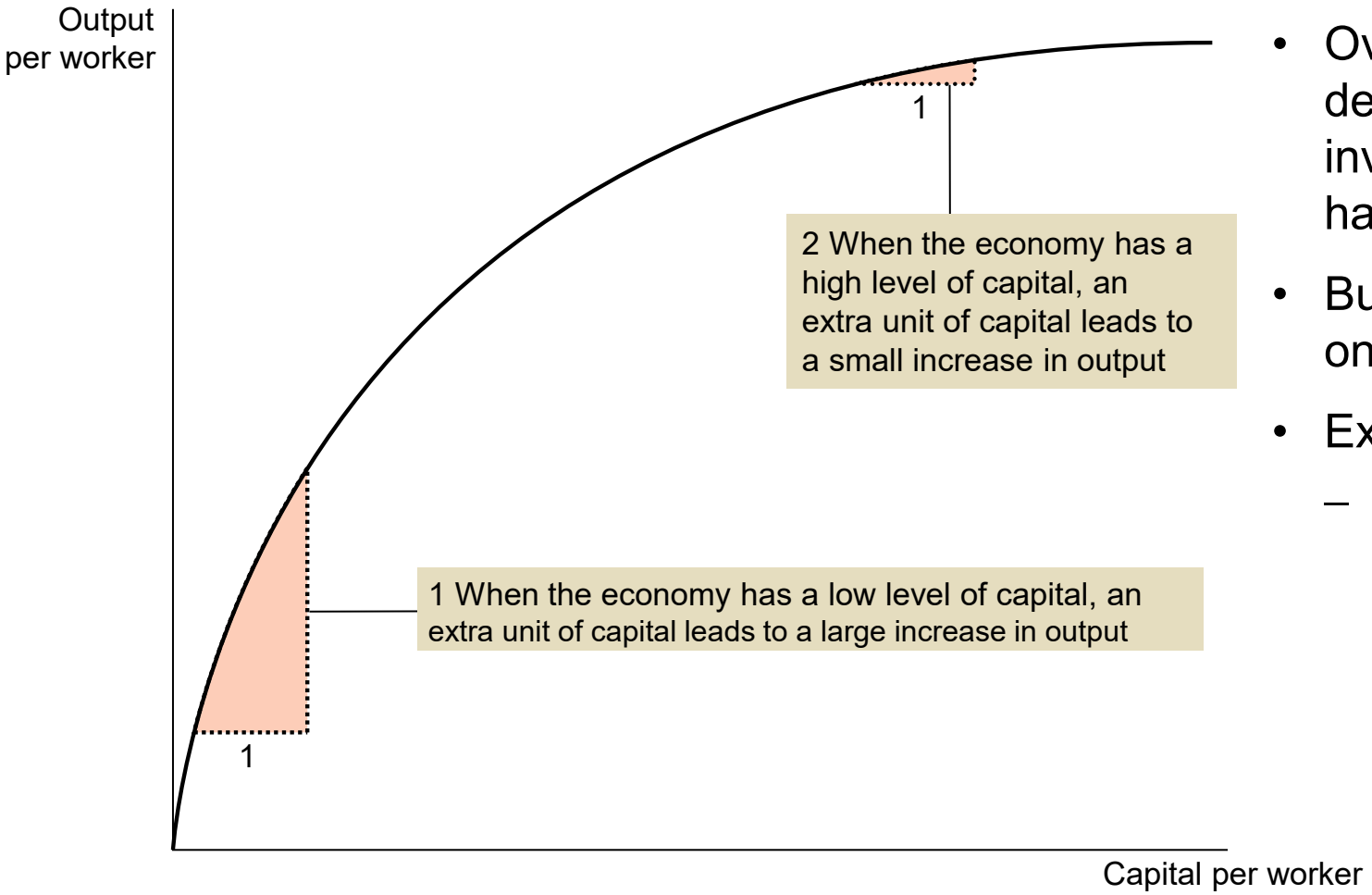
International Evidence on Investment Rates and Income per Person



Diminishing Returns and the Catch-Up Effect

- As the stock of capital rises, the extra output produced from an additional unit of capital falls; this property is called ***diminishing returns***
- Because of diminishing returns, an increase in the saving rate leads to higher growth only for a while
- In the long run, the higher saving rate leads to a higher level of productivity and income, but not to higher growth in these areas
- The ***catch-up effect*** refers to the property whereby countries that start off poor tend to grow more rapidly than countries that start off rich

Illustrating the Production Function and an example of the Catch-Up Effect



- Over 1960-1990, the US and South Korea devoted a similar share of GDP to investment, so you might expect they would have similar growth performance
- But growth was >6% in South Korea and only 2% in the US
- Explanation: the catch-up effect
  - In 1960, **K/L** was far smaller in South Korea than in the US, hence South Korea grew faster

Investment from Abroad

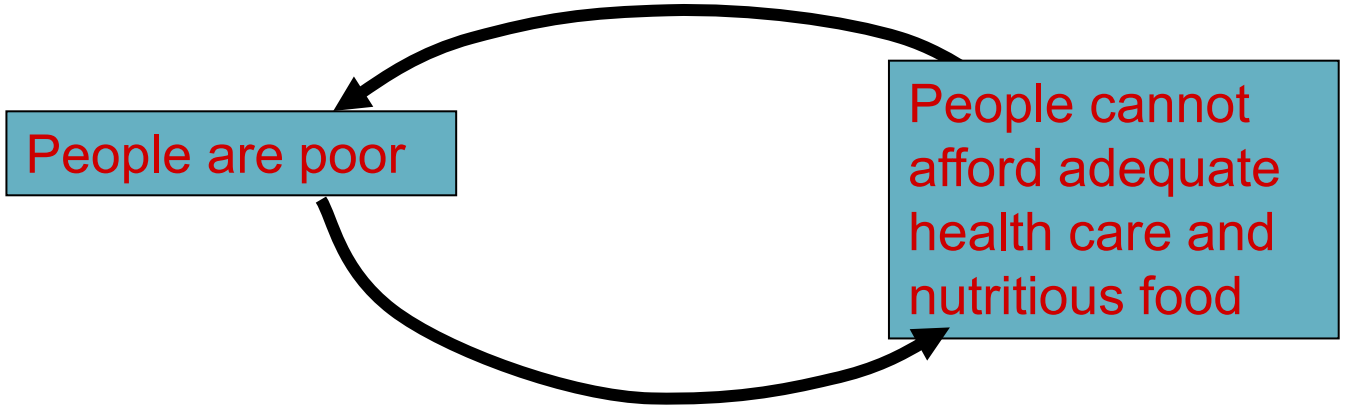
- Governments can increase capital accumulation and long-term economic growth by encouraging investment from foreign sources
- Investment from abroad takes several forms:
  - Foreign Direct Investment
    - Capital investment owned and operated by a foreign entity
  - Foreign Portfolio Investment
    - Investments financed with foreign money but operated by domestic residents
- Some of the returns from these investments flow back to the foreign countries that supplied the funds
- Especially beneficial in poor countries that cannot generate enough saving to fund investment projects themselves
- Also helps poor countries learn state-of-the-art technologies developed in other countries

Education

- For long-run growth, education is at least as important as investment in physical capital
  - Investing in human capital also involves a trade-off between the present and the future:
    - Spending a year in school requires sacrificing a year's wages now to have higher wages later
  - In the US, each year of schooling raises a person's wage, on average, by about 10 percent
- Government can enhance the standard of living by providing schools and encourage the population to take advantage of them
  - Positive externality: An educated person might generate new ideas about how best to produce goods and services, which in turn, might enter society's pool of knowledge and provide an external benefit to others
- One problem facing some poor countries is the brain drain
  - the emigration of many of the most highly educated workers to rich countries

Health and Nutrition

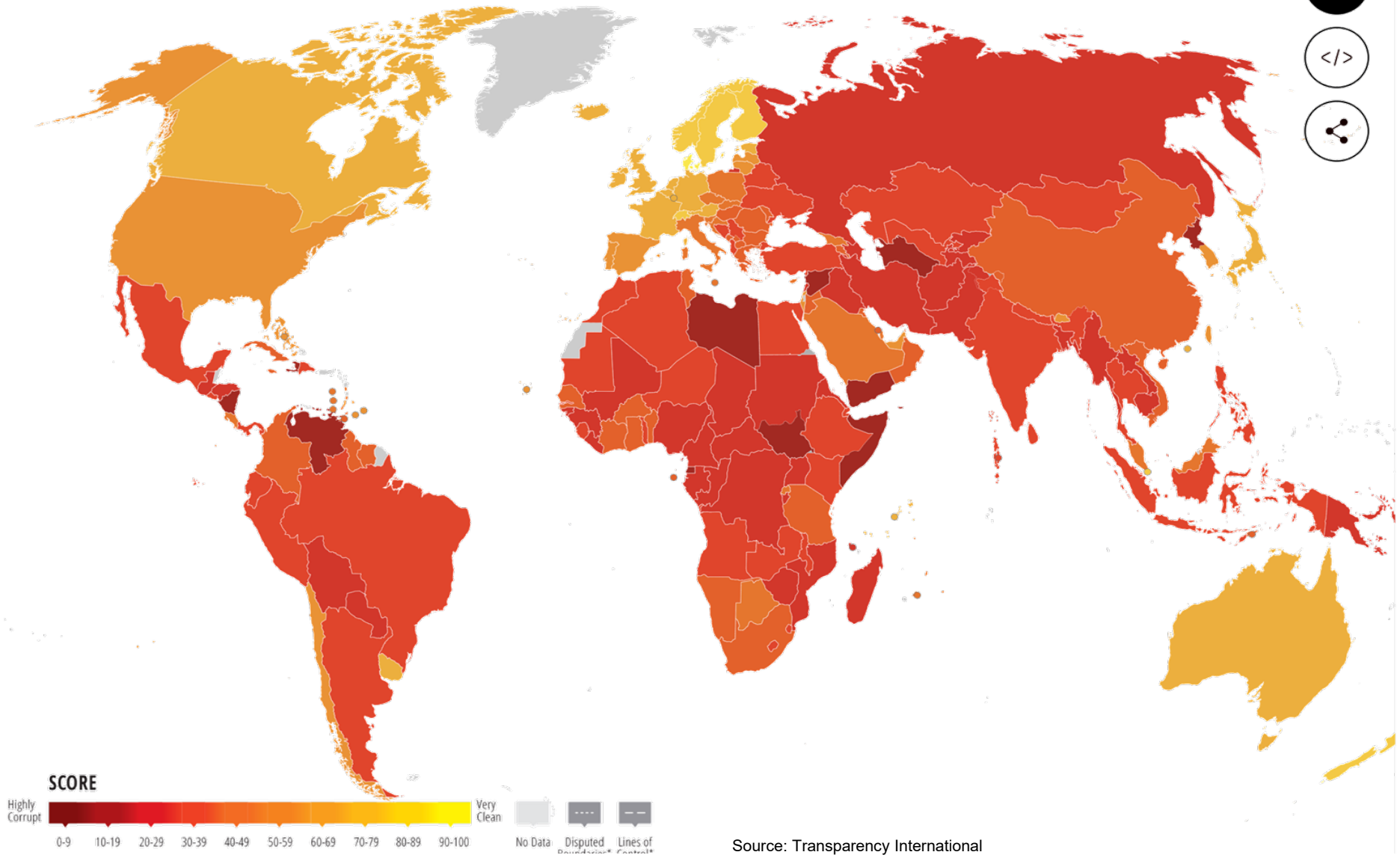
- Healthier workers are more productive
  - Health care expenditure is a type of investment in human capital
    - healthier workers are more productive
- Investments in the health of the population can lead to increase living standards
- Countries can get caught in a vicious cycle



Property Rights and Political Stability

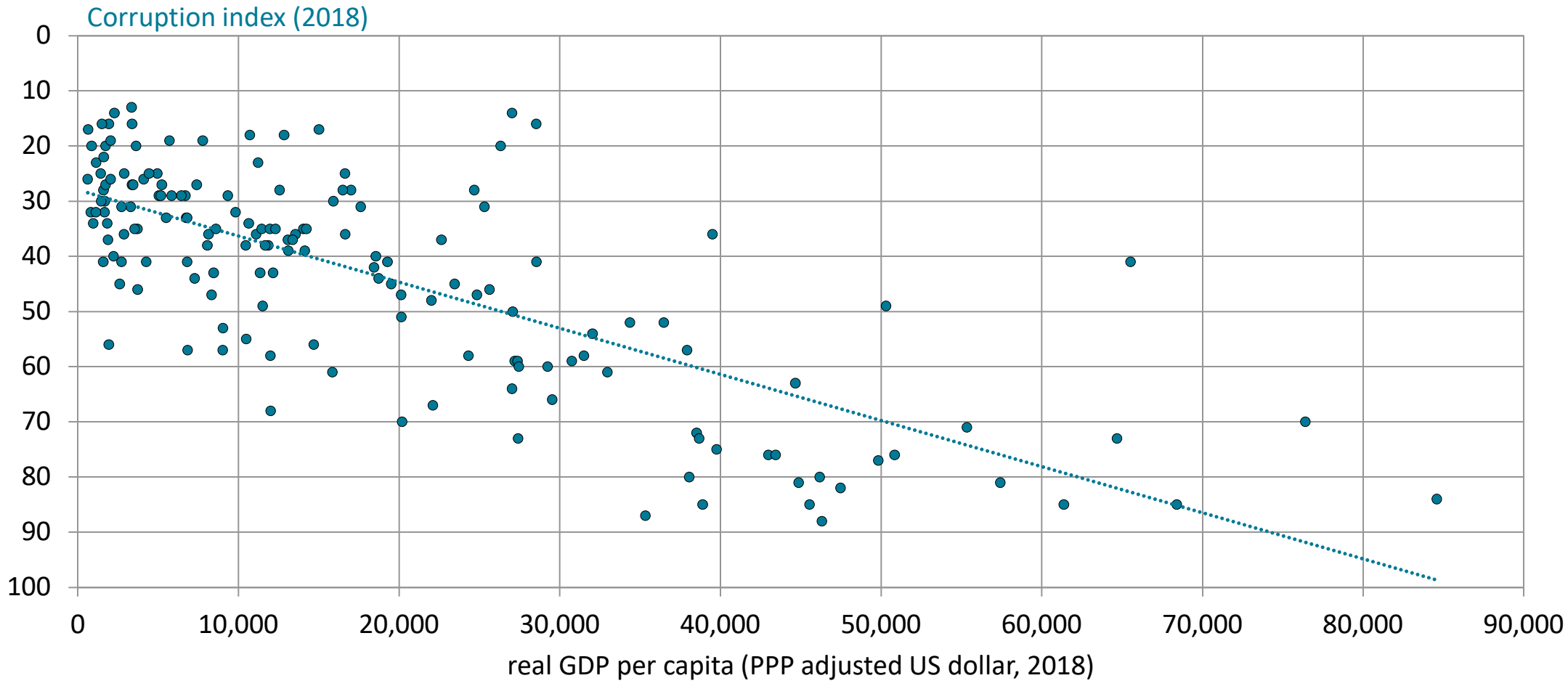
- Recall: Markets are usually a good way to organize economic activity
  - The price system allocates resources to their most efficient uses
- This requires respect for property rights
- Property rights refer to the ability to exercise authority over resources one owns
- An economy-wide respect for property rights is an important prerequisite for the price system to work
- In many poor countries, the justice system doesn't work very well:
  - Contracts aren't always enforced
  - Fraud, corruption often go unpunished
  - In some, firms must bribe government officials for permits
- It is necessary for investors to feel that their investments are secure
  - Political instability (e.g., frequent coups) creates uncertainty over whether property rights will be protected in the future

Corruption around the World (2023 Index)



Source: Transparency International

Real GDP per capita and corruption around the World (2018)



Sources: Transparency International, GGDC

Free Trade

- Trade is, in some ways, a type of technology
- A country that eliminates trade restrictions will experience the same kind of economic growth that would occur after a major technological advance
- Some countries engage in …
  - … inward-orientated trade policies, avoiding interaction with other countries
  - … outward-orientated trade policies, encouraging interaction with other countries

- Countries and regions with inward-oriented policies have generally failed to create growth
  - E.g., Argentina during the 20th century
- Countries and regions with outward-oriented policies have often succeeded
  - E.g., South Korea, Singapore, Taiwan after 1960

Technological progress - Research and Development

- The advance of technological knowledge has led to higher standards of living
- One reason is that knowledge is a public good:
  - Ideas can be shared freely, increasing the productivity of many
- Technology can offset the effects of diminishing marginal product
- Most technological advances come from private research by firms & individuals
  - Innovation and R&D are important ways in which technology growth can arise and can provide competitive advantage for firms over rivals which are both distinctive and defensible
- Government can encourage the development of new technologies through
  - Patent laws
  - Tax incentives or direct support for private sector R&D
  - Grants for basic research at universities

Population Growth

- Population growth interacts with other factors of production:
  - Stretching natural resources
    - 200 years ago, Malthus argued that population growth meant that the world was doomed to live in poverty forever
    - Since then, the world population has increased sixfold. If Malthus was right, living standards would have fallen. Instead, they've risen
    - He failed to understand that new ideas would be developed to increase the production of food and other goods, fertilizers, mechanized equipment, and new crop varieties (technological progress)
  - Diluting the capital stock
    - Lower K/L = lower productivity & living standards
    - more children = greater strain on educational system = lower H/L
  - Promoting technological progress
    - more scientists, inventors, engineers = more frequent discoveries
      - growth rates increased as the world's population increased
      - more populated regions grew faster than less populated ones

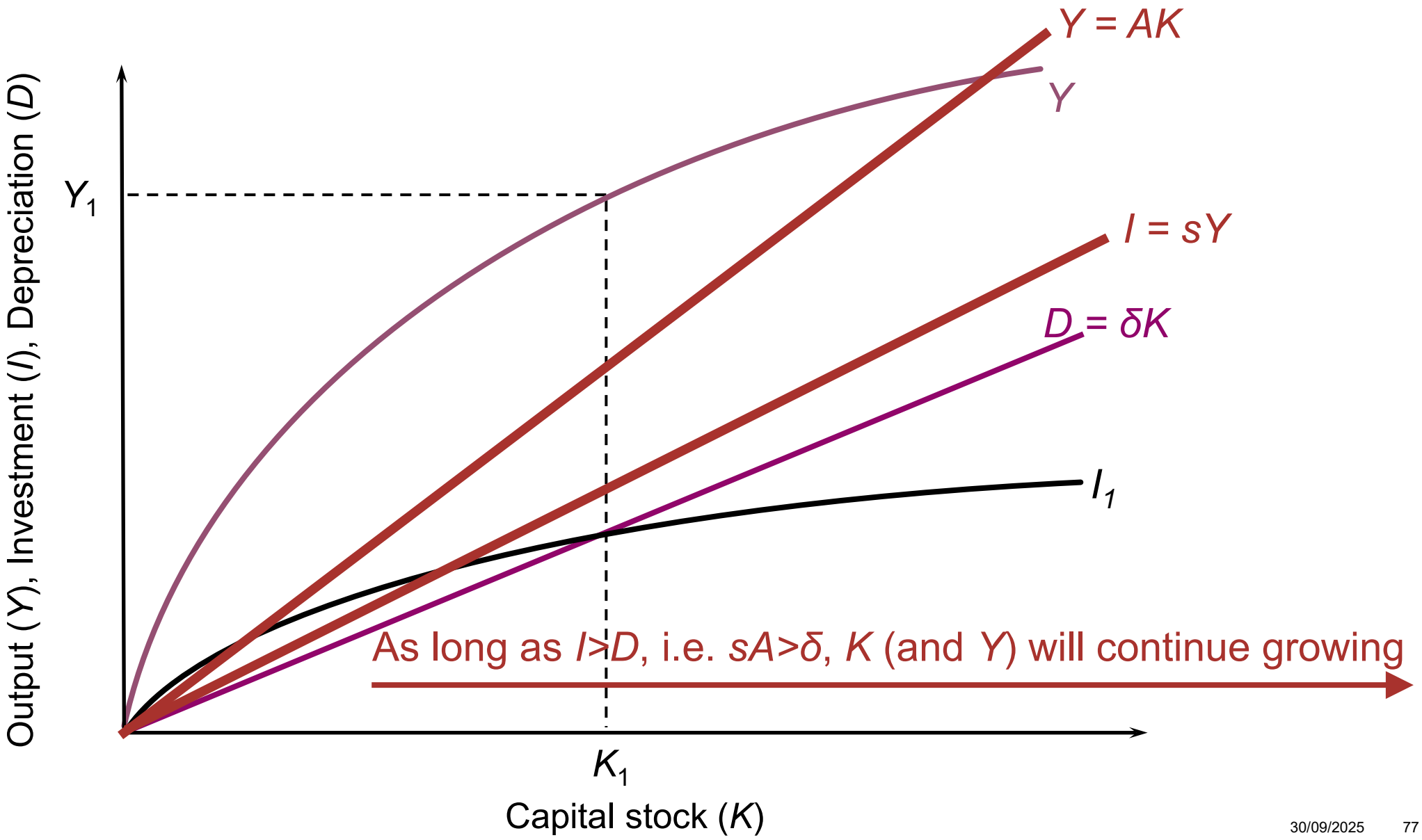
Are Natural Resources a Limit to Growth?

- Some argue that population growth is depleting the Earth's non-renewable resources, and thus will limit growth in living standards
- But technological progress often yields ways to avoid these limits:
  - Hybrid cars use less gas
  - Better insulation in homes reduces the energy required to heat or cool them
- As a resource becomes scarcer, its market price rises, which increases the incentive to conserve it and develop alternatives

Endogenous Growth Theory

- ***Endogenous growth theory*** is a theory of long-run economic growth which results from the creation of new knowledge and technology which impacts on everyone and makes them more productive as a result
- A basic endogenous growth model:
  - Production function: $Y = A K$
    where A is the amount of output for each unit of capital (A is exogenous & constant)
  - Key difference between this model & Solow:
  - Marginal product of capital is constant here, while it diminishes in the Solow model
- As in the Solow model: Investment = sY and Depreciation = δK
  - Equation of motion for total capital: $\Delta K = sY - \delta K$ or $\Delta K/K = sA - \delta$
  - If $sA > \delta$, then income will grow forever, and investment is the "engine of growth"
  - Here, the permanent growth rate depends on s – in the Solow model, it does not

A basic endogenous growth model



Summary

- Economic prosperity, as measured by real GDP per person, varies substantially around the world
- The average income of the world's richest countries is more than ten times that in the world's poorest countries
- The standard of living in an economy depends on the economy's ability to produce goods and services
- Productivity depends on the amounts of physical capital, human capital, natural resources, and technological knowledge available to workers
- Government policies can influence the economy's growth rate in many different ways
- The Solow model notes that the accumulation of capital is subject to diminishing returns
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Summary

- Productivity depends on the amounts of physical capital, human capital, natural resources, and technological knowledge available to workers
- Government policies can influence the economy's growth rate in many different ways
- The Solow model notes that the accumulation of capital is subject to diminishing returns
- Because of diminishing returns, higher saving leads to a higher growth for a period of time, but growth will eventually slow down
- Also because of diminishing returns, the return to capital is especially high in poor countries
- Endogenous growth theory explains how technology can change and offset the effects of diminishing marginal productivity. Innovation and R&D are important ways in which technology growth can arise
- Government policies can try to influence the economy's growth rate in many ways: by encouraging saving and investment, encouraging investment from abroad, fostering education, maintaining property rights and political stability, allowing free trade, promoting the research and development of new technologies, and controlling population growth